Gourav Gulia

ML Engineer • MLOps • Generative AI

Haryana, 131001 | 9588313823 | grvgulia007@gmail.com | gauravgulia | gauravgulia26

Skills

Languages	Python
Machine Learning	Scikit-Learn, TensorFlow, XGBoost, Model Evaluation, Feature Engineering
Generative AI	LangGraph, RAG, Prompt Engineering, Langchain
Computer Vision	OpenCV, FaceNet512, RetinaFace, Image Quality Assessment (PSNR, SSIM, LBP)
MLOps	Apache Airflow, MLflow, DVC, Docker, FastAPI, CI/CD
Software Eng.	REST APIs, Model Serving, Dependency Injection, Factory Pattern, Python Packaging, Modular Architecture
Data & Libraries	pandas, Pandas, NumPy, Pydantic
Databases	SQL, MongoDB, VectorDB (Milvus, FAISS, Weaviate)
Tools	Git, Linux, Streamlit

Experience

EY (Ernst & Young) — **Jun 2025 - Jun 2026**
Sr. Analyst — *Gurugram*

- Identified malpractice patterns and compliance gaps across large-scale government examination datasets using **pandas**, statistical analysis, and forensic validation techniques on multimodal data including facial images, demographic records, and system logs.
- Developed and deployed production-grade **face verification**, **image quality analysis**, and **morphing detection** pipelines for government clients including **SSC**, **HSSC**, and **NHA**, leveraging **FaceNet512**, **RetinaFace**, **PSNR**, **SSIM**, and **LBP** to improve verification accuracy across challenging image conditions.
- Designed a **weighted ensemble inference framework** optimized through **A/B Testing**, **LCB/UCB strategies**, and **ROC-based threshold tuning**, reducing false positives by nearly **11%** while improving verification consistency.
- Built an in-house **Text Similarity & Impersonation Detection System** using **Jaro-Winkler similarity**, **phonetic matching**, and **TF-IDF vectorization**, improving duplicate identity detection efficiency by approximately **25%** during forensic investigations.
- Re-architected legacy ML solutions into a modular, reusable, and API-driven framework using **FastAPI**, while implementing workflow orchestration with **Apache Airflow** and experiment tracking through **MLflow**, significantly improving deployment scalability and reproducibility.
- Optimized production inference pipelines through **multiprocessing**, **hyperparameter tuning**, and memory-efficient processing techniques including **dtype downcasting**, reducing inference latency by approximately **35%** and memory utilization by **22%**.
- Collaborated with cross-functional teams and government stakeholders to deliver enterprise **forensic AI solutions** aligned with regulatory compliance, risk governance, and standardized investigation workflows across large-scale public sector projects.

Netmax — **Oct 2024 - Feb 2025**
Jr. Data Scientist — *Chandigarh*

- Built a modular data preprocessing pipeline leveraging **MLflow**, **DVC**, and **Pandas**, which reduced data drift and noise by 20%, boosting downstream model accuracy across production workflows.

Education

M.Sc. Data Science — **Jul 2022 – May 2024**
Chandigarh University

B.Sc. Applied Science — **Jul 2019 – May 2022**
Delhi University

Featured Projects

End-to-End AI-Powered Burnout Risk Prediction | Links • • • • • **Jun 2026 - Aug 2026**

- Engineered an end-to-end **student burnout prediction system** covering data validation, feature engineering, model training, evaluation, and production inference.
- Built a configurable **ML experimentation framework** for systematic model comparison and hyperparameter optimization across multiple classification algorithms.
- Implemented a **reproducible, artifact-driven ML workflow** using DVC and MLflow for versioned datasets, experiment tracking, model management, and pipeline reproducibility.
- Productionized the model with **FastAPI, Streamlit, and Docker**, deploying a live prediction API and interactive UI with a publicly available container image.

CIP: Candidate Intelligence Platform **Jan 2026 - Mar 2026**

- Developed a **Ensembled candidate risk prediction model** using structured examination metadata and forensic indicators, providing the foundation for downstream explainable AI workflows.
- Architected a modular, configuration-driven **AI risk intelligence platform** at **EY GPS Assurance** using reusable pipeline components, artifact-based communication, **Dependency Injection**, and YAML-driven configurations, enabling scalable and maintainable ML workflows.
- Built a production-grade **MLOps pipeline** leveraging **Apache Airflow**, **DVC**, **MLflow**, **Docker**, and **FastAPI** for automated orchestration, experiment tracking, data versioning, reproducible model training, and scalable inference.
- Developed an AI-powered **Investigation Copilot** using **LangGraph**, advanced **prompt engineering**, and a **Retrieval-Augmented Generation (RAG)** architecture to retrieve SOPs, historical investigation cases, and policy documents for explainable, evidence-backed decision support.
- Designed an interactive **Streamlit** investigation dashboard integrating real-time ML predictions with conversational AI, enabling investigators to analyze candidate risk, understand model reasoning, and receive context-aware investigation recommendations through a production-ready interface.

Inspector: In-House Forensic Analytics Library for EY **Jul 2025 – Aug 2025**

- Designed an in-house modular **Python package (.whl)** and **Dockerized forensic analytics library** at **EY** for scalable image and text-based ML workflows, enabling reusable and production-ready forensic pipelines.
- Engineered a high-performance **face retrieval system** using **VectorDB** and **HNSW-based ANN indexing**, enabling near real-time similarity search across **1M+ embeddings** with optimized retrieval latency.
- Built a scalable **text forensic analytics engine** using **Jaro-Winkler similarity**, **TF-IDF vectorization**, and **nearest neighbor search** to detect impersonation and duplicate identity patterns across large-scale datasets.
- Accelerated ML pipeline performance using **multiprocessing**, **memory-efficient processing**, and advanced software engineering principles including **Dependency Injection** and **Factory Pattern**, improving maintainability and execution efficiency.

Logpunch: Custom Logging & Exception Handling Library for ML Workflows **May 2025 – Jun 2025**

- Engineered a pip-installable Python package with colored logging, real-time file logging, and detailed exception tracking, now live on **PyPI**.
- Reduced logging boilerplate by **85%** across data science pipelines via **Pydantic**- validated configs and automatic file management.
- Enabled module-aware traceback and structured error reporting—cutting debugging time by **40%** in production ML scripts.

Research Publications

Liver Disease Prediction Using Ensemble Learning

- Developed a predictive system for **liver disease detection** using multiple **machine learning algorithms**, enabling early-stage diagnosis through data-driven insights.
- Performed **comparative analysis** across models to evaluate performance using metrics such as **accuracy**, **precision**, and **recall**, achieving up to **95% accuracy**.
- Applied comprehensive **data preprocessing**, **feature selection**, and **model validation** techniques to improve generalization and robustness on medical datasets.
- Demonstrated strong understanding of **healthcare analytics**, **statistical modeling**, and **ML evaluation frameworks**, contributing to a research-backed solution for disease prediction.